Rohan Pandey

Greater Noida, Uttar Pradesh
+91 9971187301 | rohan1706pandey@gmail.com | GitHub | LinkedIn | Portfolio

Education

Bennett University — Greater Noida, Uttar Pradesh
Bachelor of Technology in Computer Science & Engineering (GPA: 8.24/10.0) — *Expected May 2027*

- **Relevant Coursework:** Data Structures & Algorithms, Database Management Systems (DBMS), Operating Systems, Object-Oriented Programming (OOP).

Experience

Full-stack Web Developer (Freelance) — Jan 2026 – Present
The Design Network (Startup) — *Remote*

- Deployed the company's primary landing page from scratch using **React**, focusing on a high-performance, SEO-optimized frontend to drive client acquisition and brand presence.
- Engineered a **WebAR** visualization tool using **Three.js** for intelligent wall mapping, spatial calculation algorithms, allowing users to overlay 3D interior units and furniture onto real-world surfaces directly within the browser.

Machine Learning Engineer Intern — Jun 2025 – Jul 2025
Keenheads — *Greater Noida, Uttar Pradesh*

- Optimized a LeNet based ML inference pipelines on FPGA hardware using HLS4ML and Vivado.
- Achieved **94.84% model compression**, significantly reducing latency and hardware resource usage for edge deployment.
- Analyzed system bottlenecks and fine-tuned compression parameters to ensure high-performance data processing.

Projects

PictoPy (Open Source Contributor) — July 2025 – Present

- Architecting a local-first data pipeline utilizing **FastAPI** and **Rust** for high-performance media ingestion.
- Implemented **unit tests** for database schemas, timeline scrollbar component to efficiently store and query complex metadata and file relationships.
- Integrated vector embeddings (FaceNet) and clustering algorithms (DBSCAN) to structure unstructured image data.
- Optimized inference speed using ONNX Runtime, enabling real-time analysis on local hardware.

Arogya (Physiotherapy Aid) — Jan 2025 – July 2025

- Built a precision joint movement tracking system integrating ESP32 and 3-axis accelerometer with a Unity-based simulation.
- Processed real-time sensor data streams to track joint angles and provide instant user feedback.

Drive Guardian (Autonomous Systems) — Jan 2024 – Oct 2025

- Developed an automated braking algorithm capable of real-time object detection and lane recognition.
- Calculated dynamic safety metrics to automate decision-making processes based on sensor input feeds.
- Engineered a robust computer vision pipeline to process video streams for obstacle avoidance.

Technical Skills

Languages: Python, C++, SQL, JavaScript, TypeScript, Rust
Frontend: React, Next.js, Tailwind CSS, Redux
Backend & Data: FastAPI, Flask, SQLite, MySQL, Node.js, Docker
Machine Learning: PyTorch, TensorFlow, Pandas, NumPy, OpenCV, YOLO, ONNX
Tools & Platforms: Git, GitHub, AWS, Linux, HLS4ML, Vivado

Leadership

Vice Chairperson — Sep 2025 – Present
Data Science Society — *Bennett University*

- Collaborated with Google Crowdsource to organize a technical workshop for 200+ students.
- Led technical initiatives and launched the official GitHub organization to foster open-source contributions.